

Euclid's Algorithm and the RSA Encryption Scheme

- For each pair of integers (a, b) below, find $d = \gcd(a, b)$ **and** find a pair of numbers j and k such that $d = j \cdot a + k \cdot b$. (Use the Extended Euclidean algorithm.)
 - 90 and 56
 - 91 and 89
 - 711 and 75
 - 815 and 75
 - 112 and 196
 - 366 and 150
- For each triple of integers p, q, e given below, find the (smallest positive) value of d that will work for the private key of the RSA algorithm (where, as usual, $n = p \cdot q$ and $\phi(n) = (p - 1)(q - 1)$, and $ed \equiv 1 \pmod{\phi(n)}$).
Then suppose that the message (integer) $M = 25$ is to be encoded using the public key n, e pair. Find the ciphertext that corresponds to that message M .
 - $p = 7, q = 13, e = 5$
 - $p = 5, q = 11, e = 3$
 - $p = 7, q = 17, e = 7$
- Finally, suppose that $p = 5, q = 17$, and $e = 13$. First find the private key d for the RSA method with these parameters. Then decrypt the ciphertext messages, C, below to find the original (plaintext) messages.
 - 12
 - 9
 - 27
 - 84

Solution:

- First let us recall the Extended Euclidean algorithm. The pseudocode for this method is given below:

EXTENDED EUCLIDGCD(a, b)

Input: Nonnegative integers a and b (not both zero).

Output: $d = \gcd(a, b)$, integers j, k where $d = j \cdot a + k \cdot b$.

- if** $b = 0$ **then**
- return $(a, 1, 0)$
- $r \leftarrow a \bmod b$
- Let q be the integer such that $a = q \cdot b + r$ (that is, $q = \lfloor \frac{a}{b} \rfloor$).
- $(d, j, k) \leftarrow \text{ExtendedEuclidGCD}(b, r)$
- return $(d, k, j - kq)$

We can use this algorithm for each pair of numbers in Question 1. As this is a recursive procedure, we fill in the table below from left to right for the rows involving a, b, q , and r , and then we fill it in from right to left for the rows involving j and k . We will assume that $a \geq b$ (as will always be the case after the first step of the algorithm, so let's just assume this is the case).

(a)

a	90	56	34	22	12	10	2
b	56	34	22	12	10	2	0
q	1	1	1	1	1	5	-
r	34	22	12	10	2	0	-
j							1
k							0

We've first established that $\gcd(90, 56) = 2$.

Having made the recursive function calls, we're now able to fill in the values of j and k from right to left. The last column (as shown) will always be 1 and 0 (assuming that a and b are not both 0, of course). To get the values of j and k in a column, we use the values of j and k from the previous column (the one to its right, as we're filling in the table from right to left) and the value of q that is in the *same column* that we're currently filling in.

So the final table looks like this:

a	90	56	34	22	12	10	2
b	56	34	22	12	10	2	0
q	1	1	1	1	1	5	-
r	34	22	12	10	2	0	-
j	5	-3	2	-1	1	0	1
k	-8	5	-3	2	-1	1	0

Thus we have $\gcd(90, 56) = 2$ and $2 = (5) \cdot 90 + (-8) \cdot 56$.

You can double check your work at each step, as for *each* column, using those values of a, b, j , and k , you should always get a linear combination that equals the GCD. For example, we have

$$2 = (-1) \cdot 22 + (2) \cdot 10$$

and also

$$2 = (2) \cdot 34 + (-3)22.$$

For the remainder of the pairs, we'll only give the final table.

(b)

a	91	89	2	1
b	89	2	1	0
q	1	44	2	-
r	2	1	0	-
j	-44	1	0	1
k	45	-44	1	0

Hence, $\gcd(91, 89) = 1$ and $1 = (-44) \cdot 91 + (45) \cdot 89$.

(c)

a	711	75	36	3
b	75	36	3	0
q	9	2	12	-
r	36	3	0	-
j	-2	1	0	1
k	19	-2	1	0

So $\gcd(711, 75) = 3$ and $3 = (-2) \cdot 711 + (19) \cdot 75$.

(d)

a	815	75	65	10	5
b	75	65	10	5	0
q	10	1	6	2	-
r	65	10	5	0	-
j	7	-6	1	0	1
k	-76	7	-6	1	0

Hence $\gcd(815, 75) = 5$, and we have $5 = (7) \cdot 815 + (-76) \cdot 75$.

(e)

a	196	112	84	1
b	112	84	28	0
q	1	1	3	-
r	84	28	0	-
j	-1	1	0	1
k	2	-1	1	0

So we have $\gcd(196, 112) = 28$ and $28 = (-1) \cdot 196 + (2) \cdot 112$.

(f)

a	366	150	66	18	12	6
b	150	66	18	12	6	0
q	2	2	3	1	2	-
r	66	18	12	6	0	-
j	-9	4	-1	1	0	1
k	22	-9	4	-1	1	0

Finally we see $\gcd(366, 150) = 6$ where $6 = (-9) \cdot 366 + (22) \cdot 150$.

2. For these problems to find d , we can again use the Extended Euclidean algorithm. The main idea is that given e and $\phi(n)$ (where $\gcd(e, \phi(n)) = 1$), we can first find the integers j and k such that $j \cdot \phi(n) + k \cdot e = 1$.

Then having found j and k , we can reduce this equation modulo $\phi(n)$, so

$$[j \cdot \phi(n) + k \cdot e] \pmod{\phi(n)} \equiv 1 \pmod{\phi(n)}.$$

But we have

$$[j \cdot \phi(n) + k \cdot e] \pmod{\phi(n)} \equiv j \cdot \phi(n) \pmod{\phi(n)} + k \cdot e \pmod{\phi(n)},$$

and

$$j \cdot \phi(n) \equiv 0 \pmod{\phi(n)} \text{ as } j \cdot \phi(n) \text{ is divisible by } \phi(n).$$

Thus we conclude that $k \cdot e \equiv 1 \pmod{\phi(n)}$.

Now if $k > 0$ then we can take $d = k$ for the RSA decryption method. However, if $k < 0$, then we will add a multiple of $\phi(n)$ to k until we obtain a positive number and this will be the value of d that we use. Some examples should help to clear this up.

(a) $p = 7, q = 13, e = 5$

In this case we have $n = p \cdot q = 91$ and $\phi(91) = (p - 1)(q - 1) = 72$. Then we use the Extended Euclidean algorithm.

a	72	5	2	1
b	5	2	1	0
q	14	2	2	-
r	2	1	0	-
j	-2	1	0	1
k	29	-2	1	0

So we have $(-2) \cdot 72 + (29) \cdot 5 = 1$ implying that $29 \cdot 5 \equiv 1 \pmod{72}$. In this case we can take $d = 29$.

Then we want to compute $25^5 \pmod{91}$ to encode the given message. First we find that $25^2 \pmod{91} = 625 \pmod{91} = 79$. Then

$$25^4 \pmod{91} = 79^2 \pmod{91} = 6241 \pmod{91} = 53.$$

Finally $25^5 \pmod{91} = 53 \cdot 25 \pmod{91} = 1325 \pmod{91} = 51$. So the ciphertext message that corresponds to $M = 25$ is $C = 51$.

(You could check that $51^{29} \pmod{91} = 25$ but you'd want to use the "fast exponentiation algorithm" that we talked about in class to do this by hand.)

(b) $p = 5, q = 11, e = 3$

We do the same as before, with $n = 55$ and $\phi(55) = 40$.

a	40	3	1
b	3	1	0
q	13	3	-
r	1	0	-
j	1	0	1
k	-13	1	0

This time we have that $(1) \cdot 40 + (-13) \cdot 3 = 1$. This means that $(-13) \cdot 3 \equiv 1 \pmod{40}$.

Using what we said earlier, to get our value of d , we will add a multiple of 40 to the number -13 until we get a positive number. Namely, we take $d = -13 + 40 = 27$.

Note that we still have the true congruence $27 \cdot 3 \equiv 1 \pmod{40}$ since $27 \cdot 3 - 1 = 80$ is a multiple of 40.

To encode the message $M = 25$ we want to find $25^3 \pmod{55}$. Then $25^2 \pmod{55} = 20$, and so $25^3 \pmod{55} = 25 \cdot 20 \pmod{55} = 5$.

The message $M = 25$ is encoded as the ciphertext $C = 5$ in this case.

(c) $p = 7, q = 17, e = 7$

The procedure is clear now (I hope!). With $n = 7 \cdot 17 = 119$ and $\phi(119) = 6 \cdot 16 = 96$, we first find d .

a	96	7	5	2	1
b	7	5	2	1	0
q	13	1	2	2	-
r	5	2	1	0	-
j	3	-2	1	0	1
k	-41	3	-2	1	0

So as usual we have $(3) \cdot 96 + (-41) \cdot 7 = 1$.

Thus $(-41) \cdot 7 \equiv 1 \pmod{96}$. So we take $d = -41 + 96 = 55$.

The message $M = 25$ is encrypted as $C = 25^7 \bmod 119 = 32$.

3. Here we have $p = 5, q = 17, e = 13$. Thus $n = 5 \cdot 17 = 85$ and $\phi(85) = 4 \cdot 16 = 64$. You know the drill by now to find the decryption key d .

a	64	13	12	1
b	13	12	1	0
q	4	1	12	-
r	12	1	0	-
j	-1	1	0	1
k	5	-1	1	0

So $(-1) \cdot 64 + (5) \cdot 13 = 1$ and therefore $5 \cdot 13 \equiv 1 \pmod{64}$. In this case we have $d = 5$.

Using this decryption key, we can decode the messages that we're given by finding $C^d \bmod 85 = C^5 \bmod 85$.

(a) $12^5 \bmod 85 = 37$.

(b) $9^5 \bmod 85 = 59$.

(c) $27^5 \bmod 85 = 57$.

(d) $84^5 \bmod 85 = 84$.

(So 84 is "decoded" as the number 84, and hence, 84 is also encoded as 84.)

For all of these numbers, we can also check that $M^{13} \bmod 85 = C$. For example, we have $37^{13} \bmod 85 = 12$ (a small verification that the RSA scheme works as intended).